Optimizing Inventory and Sales Forecasting at Pastry palace: A Data-Driven Approach to be Efficient and Organised

A Mid-Term report for the BDM capstone Project

Submitted by

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1. Executive Summary

This project focuses on **Pastry Palace**, a small-scale food retail snacks and bakery shop located in *Indira Complex, Central Hospital Road, Dakra, Khalari, Churi, Jharkhand 829210*. The business has been running since **August 2021** and operates in a **semi-rural area**, serving local walk-in customers. It offers a variety of freshly made items such as **pizza, momos, handmade cakes, and biscuits**, along with selected **cold drinks and packaged FMCG products**. While the business decisions rely on the owner's experience, it currently lacks the tracking and managing of **sales and stocks** of **perishable and non-perishable items**.

The dataset was manually created after repeated store visits and discussions with the owner, covering sales and stock behavior for **June 2025**. The data includes over **800 entries across 12 columns** including **quantity sold, quantity prepared, profit, and item category**. Descriptive statistics such as **mean, median, standard deviation, percentiles, and mode** were used to summarize the data. These help explain variations in product demand and highlight key issues in stock and profit management.

This project follows a structured approach using Excel to identify trends, wastage, and sales behavior through tools like pivot tables, ABC analysis, and profit margin comparisons. Descriptive statistics revealed that while ***Essentials*** and ***Packaged*** items had the highest average sales, *Fresh* items like **Paneer Pizza** and **Chocolate Cake** generated the most profit. High wastage was observed in items like **Pav Bhaji** and **Ice Cream**, and Friday sales consistently dipped due to local vegetable market activity. These early insights support the core problem areas and serve as a foundation to drive data-backed improvements in stock planning, demand alignment, and profitability optimization.

2.Proof of Originality-

2.1 Interaction Video - shop video.mp4

2.2 Shop Image - shop image

2.3 Signed Letter - letter head.PDF

3. Metadata – The data collected here is primarily based on verbal responses and notes gathered by visiting the store regularly, often with a gap of 2–3 days or a week, over a period of one month (from **01-06-2025 to 30-06-2025**). The entries reflect the owner's best understanding of daily item sales, wastage, and pricing during this time.

The final dataset contains **12 columns and 906 rows**, with each row representing data for a specific item on a particular date. Below is a detailed description of each variable, including its unit, data type, and its purpose in the analysis.

Variable Name	**Unit**	**Data Type**	**Description**
Date	DD-MM-YYYY	Date	Day on which data was recorded. Helps identify daily trends and event impact.
Item Name	-	Categorical	Name of the product sold (e.g., "Pizza - Paneer", "Ice Cream - Amul").
Category	-	Categorical	Classifies products as Fresh, Packaged, Essentials, or Birthday Items.
Initial Stocks	Units	Integer	Inventory carried forward from the previous day. Tracks daily availability.
QuantityPrepared	Units	Integer	Number of items made per day (only applicable for fresh products).
Quantity Sold	Units	Integer	Number of items sold per day.
Closing Stock	Units	Integer	Remaining stock at the end of the day.
Wastage/Remaining	Units	Integer	Unused or spoiled items. Applicable mainly to perishables.
Selling Price(₹)	₹ (INR)	Float	Estimated cost of purchase or preparation per unit.

Cost Price (₹)	₹ (INR)	Float	Estimated cost of purchase or preparation per unit.
Profit per Unit (₹)	₹ (INR)	Float	Selling Price – Cost Price. Indicates unit-level profitability.
Total Profit (₹)	₹ (INR)	Float	Profit per unit × Quantity Sold. Reflects total contribution to revenue.

Snapshot of Metadata - Pastry Palace Data

Date	Item Name	Category	Initial Stocks	Quantity Prepared	Quantity Sold	Closing stock	Wastage/Remaining	Selling Price (₹)	Cost Price (₹)	Profit per Unit (₹)	Total Profit (₹)	Notes
1-Jun-2025	Birthday - Balloons	Birthday Items	42		4	38		5	3	2	8	
1-Jun-2025	Birthday - Candles	Birthday Items	17		1	16		10	7	3	3	
1-Jun-2025	Birthday - Caps	Birthday Items	21		3	18		10	6	4	12	
1-Jun-2025	Birthday - Poppers	Birthday Items	10		1	9		20	13	7	7	
1-Jun-2025	Biscuit - Jeera	Packaged	60		11	49		20	15	5	55	
1-Jun-2025	Biscuits - Cream	Packaged	47		8	39		20	15	5	40	
1-Jun-2025	Bread	Essentials	14		9	5	0	25	18	7	63	
1-Jun-2025	Burger	Fresh		11	11	0	0	50	33	17	187	
1-Jun-2025	Cake - Butterscotch	Fresh	3	2	1	4	0	250	138	112	112	School Reopening Day boost/wastage power cut-rain
1-Jun-2025	Cake - Chocolate	Fresh	4	2	3	3	0	250	138	112	336	School Reopening Day boost
1-Jun-2025	Chocolate - Silk	Packaged	25		2	23		70	63	7	14	
1-Jun-2025	Chocolate - ₹10 Bar	Packaged	40		8	32		10	9	1	8	

Fig - 3.1

4. Descriptive statistics

Variables	Quantity sold	Quantity Prepared	Wastage/Remaining	Profit per Unit (₹)	Total Profit (₹)
count	905	428	498	905	905
Average	4.63	4.23	0.09	20.08	58.73
Median	4	4	0	8	40
StandardDeviation	2.93	3.21	0.47	28.52	64.00
Minimum	0	0	0	0	0
25th Percentile	2	2	0	5	18
75th Percentile	6	6	0	17	80

Maximum	15	16	7	112	672
Mode	2	2	0	7	0
Sum	4196	1812	47	18173	53159

Table - 4.2

A descriptive statistical summary was computed for five key variables: **Quantity Sold**, **Quantity Prepared**, **Wastage/Remaining**, **Profit per Unit (₹)**, and **Total Profit (₹)**. These metrics offer a foundational understanding of performance trends across June 2025 for Pastry Palace's product sales.

- **Quantity Sold** averaged **4.64 units** per item per day, with a median of 4 and a mode of 2. This indicates a moderate but consistent demand for most products.
- **Quantity Prepared** had a slightly lower average of **4.23 units**, suggesting fairly close alignment between production and actual sales, minimizing over-preparation.
- **Wastage/Remaining** remained very low, with a mean of **0.09 units**, showing efficient stock utilization for most fresh items.
- **Profit per Unit** displayed a high spread (**mean: ₹20.08; std. dev: ₹28.52**) due to the wide variety of items — from ₹5 chocolates to premium cakes and ice creams.
- **Total Profit per Item** averaged **₹58.74**, but was highly skewed with a maximum observed profit of **₹672**, showing that a few items heavily influence revenue.

Percentile values offered additional insights into distribution:

- **25th percentile** of total profit: ₹18
- **75th percentile** of total profit: ₹80
- This indicates that while most products remain within a moderate profit range, a small number of items outperform significantly.

The **mode** values (mostly 0 or 2) highlight that low-selling or low-margin items are common across the dataset. These often represent backup or seasonal products with irregular demand.

These descriptive statistics validate the core problem areas raised in the project:

- The near match between preparation and sales supports the owner's intuitive planning, but also suggests that data could now fine-tune this further.
- High variability in per-unit profits signals an opportunity to **optimize pricing and production around high-margin products**, a goal central to this study.
- The low but present wastage flags specific items (like **Pav Bhaji** or **Pastries**) for improvement in forecasting.

Such statistics help translate gut-feeling-based decisions into measurable patterns, aligning with the project's aim to make operations more data-driven and efficient.

To explore demand patterns, a pivot table was created to show **average daily quantity sold by product category**:

Category	AVERAGE of Quantity Sold
Birthday Items	2.32
Essentials	5.84
Fresh	4.08
Packaged	5.82
Grand Total	**4.636464088**

Essentials (e.g., milk, bread) and Packaged items (e.g., chocolates, biscuits) have the **highest daily movement**, reflecting consistent customer demand. Fresh products like pizza and cake show moderate sales but typically have **higher profit margins**. Birthday items are more **event-driven** and contribute less to daily turnover.

Table 4.1

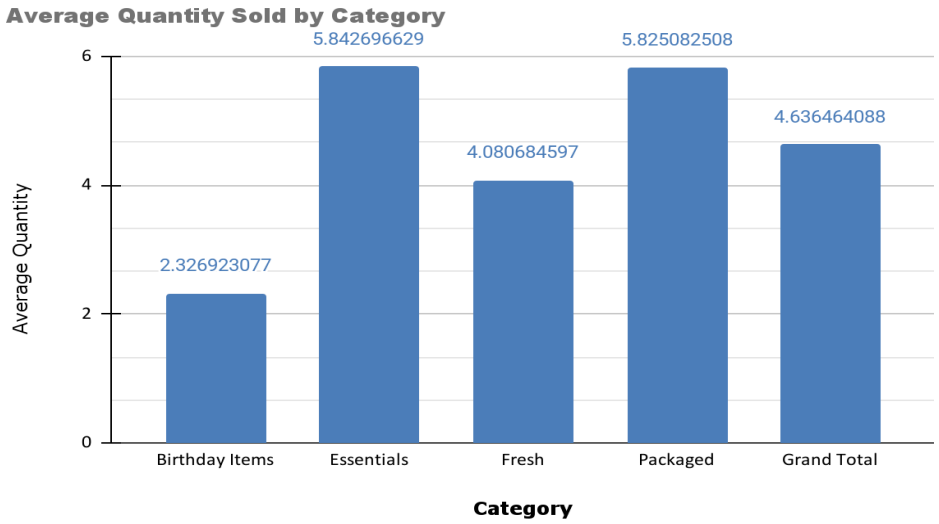

Chart 4.1: Average Quantity Sold by Product Category

- Chart 4.1 demonstrates that Essentials and Packaged products have the highest average daily sales, followed by Fresh items. Birthday Items show the lowest daily movement, highlighting their event-specific demand. This supports the idea that stocking decisions should be prioritized around frequently sold categories to optimize inventory.

5. Detailed Explanation of Analysis Process/Method

To address the key business challenges—such as the absence of structured stock tracking, difficulty in identifying high-performing products, and unpredictable wastage—a step-by-step analysis approach was implemented using Google Sheets. The overall methodology was designed to align with the real-time operational flow of Pastry Palace and to generate actionable insights with minimal technical complexity.

5.1 Data Collection and Structuring

- The dataset was manually built through consistent store visits and verbal estimations from the owner for the month of **June 2025**.
- Each entry records a specific item's behavior on a specific date.
- A total of **906 rows** and **12 columns** were compiled, covering details such as:
  - ➔ **Date, Item Name, Category, Initial Stock, Quantity Prepared, Quantity Sold, Wastage, Closing Stock, Selling Price, Cost Price, Profit per Unit, Total Profit**.
- **Closing Stock** = Initial Stock + Quantity Prepared – Quantity Sold
- **Profit per Unit** = Selling Price – Cost Price
- **Total Profit** = Profit per Unit × Quantity Sold

These metrics were essential to understand product-level sales behavior, profitability, and stock dynamics.

5.2 Descriptive Statistical Analysis

Descriptive statistics were computed for the following variables: **Quantity Sold, Quantity Prepared, Wastage, Profit per Unit, and Total Profit**. The following measures were calculated using spreadsheet formulas:

- **Central Tendencies**: Mean, Median, Mode
- **Dispersion**: Standard Deviation, 25th & 75th Percentiles, Min and Max
- **Volume**: Count, Sum

These helped create a baseline understanding of which products were consistent performers, which showed variability, and which needed attention for profitability or stocking.

5.3 Visual & Pivot-Based Insights

Multiple **pivot tables and charts** were developed to visualize category and time-based trends:

- **Bar Chart**: Average Quantity Sold per Category (Essentials and Packaged had highest average sales).
- **Pareto Chart / Bar Graph**: Top profit-contributing items (Paneer Pizza, Chocolate Cake, Burgers).
- **Line Chart**: Daily Profit Trend (highlighted dips on Fridays and spikes around Father's Day).
- **Bar Chart**: Fresh Items with Highest Wastage (e.g., Pav Bhaji, Ice Cream).

These visuals directly supported identification of bestsellers, slow movers, and event-driven spikes or drops.

5.4 Analytical Techniques Justification

Each analytical method was chosen to match the business objective of improving **inventory control** and **profitability**:

ABC Analysis: Prioritized products based on their total profit contribution to help optimize inventory decisions and focus on high-impact items.

Profit Margin Comparison: Highlighted items that seemed popular but had low actual margins, helping the owner focus on what truly drives profit.

Wastage Tracking: Helped detect frequent over-preparation of fresh/perishable items like Pav Bhaji and pastries, reducing losses.

Stock Movement Alerts: Identified fast-moving products that often ran out of stock, prompting the need for timely restocking.

5.5 Tool Justification

- **Google Sheets** was chosen for its simplicity, flexibility, and ability to handle real-time updates and shareable visualizations.
- All calculations, pivot tables, and graphs were created using **in-built spreadsheet formulas and chart tools**.

6. Results and Findings

The following findings are based on the dataset created through field visits, discussions with the shop owner, and manual estimations. Due to the informal nature of recordkeeping at Pastry Palace, only products with reasonably reliable information regarding sales volume, pricing, and wastage were included. Very low-selling or low-margin items were excluded, as sufficient data was unavailable.

Each chart and analysis presented below gives insights into product performance, inventory patterns, and overall profitability.

6.1 - Top 10 Profit-Making Products

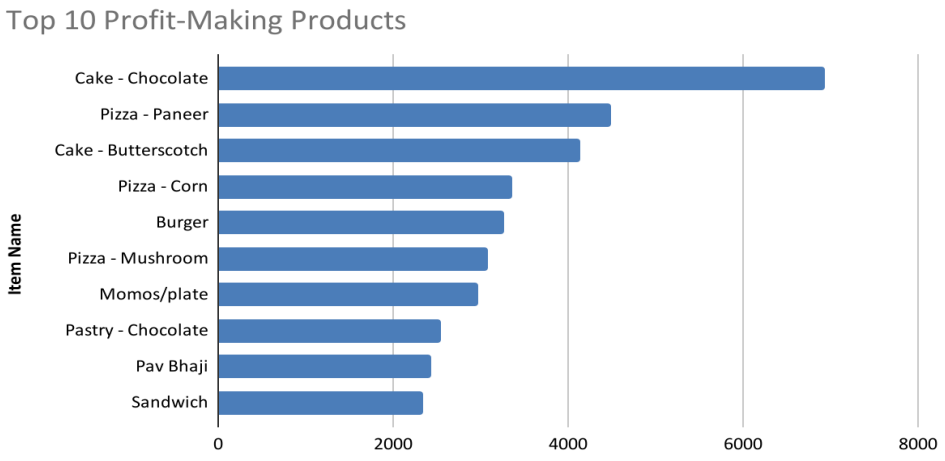

Chart 6.1: Highest Total Profit by Product

- Chocolate Cake, Paneer Pizza, and Burgers emerge as top profit contributors. These products offer both healthy margins and consistent demand. Prioritizing these items in terms of shelf visibility and preparation planning can help the business improve profitability.

6.2 - Items with Highest Wastage (Fresh Items Only)

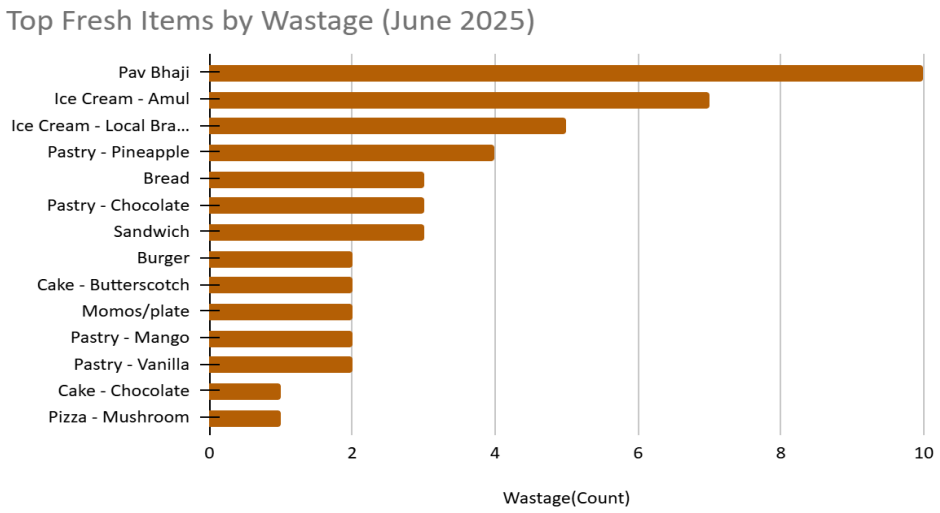

Chart 6.2: Top Fresh Items by Wastage

- Items like Pav Bhaji and Ice Creams had the most wastage during June. This could be due to over-preparation or mismatch between expected and actual demand or power cut. Tracking this consistently can help the store reduce unnecessary losses and plan better quantities per day.

6.3 - Daily Sales Trend Over Time

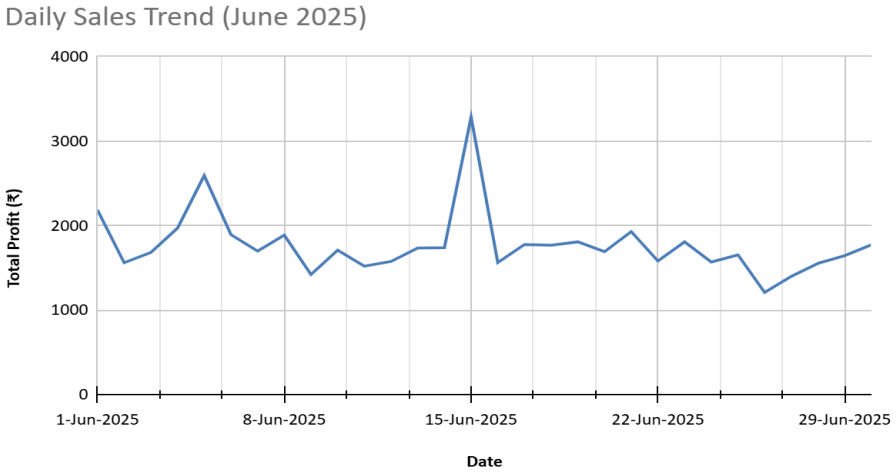

Chart 6.3: Daily Sales Trend Across June 2025

- The line chart shows fluctuations in total profit across the month. Visible dips on Fridays confirm lower footfall due to the local sabzi market. Spikes around 15 june due to father's day suggest increased purchases of cakes and chocolates. This trend helps plan better for events and slow days.

6.4 - Total Profit by Product Category

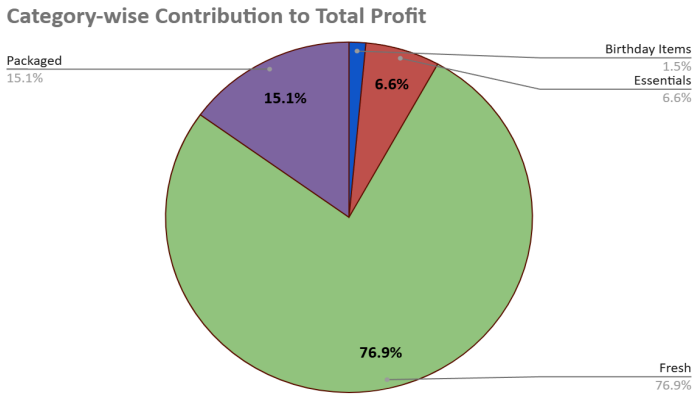

Chart 6.4: Category-wise Contribution to Total Profit

- The chart shows that **Fresh items** like pizza and cakes are the main profit drivers, followed by **Packaged products** such as biscuits, chocolates, and ice creams. Essentials and Birthday Items contribute less comparatively. This insight helps the owner focus on high-margin, high-performing product categories for future growth.

Summary of Key Findings

- **High Wastage**: Items like Pav Bhaji had frequent wastage, suggesting the need to adjust preparation based on actual demand.
- **Profit Drivers**: Fresh and Packaged items are the most profitable and should be prioritized.
- **Event Sensitivity**: Sales increase during events like Father's Day and drop on Fridays due to local market competition.
- **Profit Variability**: There is a wide range in profit per unit across items, highlighting the importance of margin-based planning.
- **Data Limitations**: A few low-selling or low-margin items were excluded, as the owner couldn't provide sufficient information about them.